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Dekan/Dean



FAKULTAS TEKNIK/FACULTY OF ENGINEERING

Program Studi : Teknik Elektro - Teknik Kimia - Teknik Industri

PSDKU Rekayasa Industri - Informatika Profesi Insinyur - Teknik Kimia (S2)

Study Program : Electrical Engineering - Chemical Engineering - Industrial Engineering

Industrial Engineering (PSDKU) - Informatics - Professional Engineer Program

Chemical Engineering (Master's)

FAKULTAS TEKNIK

FAKULTAS TEKNIK

VISI

Fakultas Teknik UKWMS menjadi pusat pendidikan teknik yang terbuka, diakui di tingkat nasional hingga internasional dalam bidang teknologi industri dan informasi, untuk berkontribusi dan berdampak pada kesejahteraan masyarakat dengan dijiwai oleh nilai-nilai Pancasila dan prinsip-prinsip Katolik.

MISI

1. Menyelenggarakan pendidikan untuk menghasilkan lulusan yang berintegritas, kompeten, dan profesional dalam bidang teknologi industri dan informasi.
2. Melaksanakan penelitian dan pengabdian kepada masyarakat dalam bidang teknologi industri dan informasi untuk meningkatkan kesejahteraan masyarakat.
3. Memperluas jejaring kerja sama pentahelix untuk pengembangan Tridharma Perguruan Tinggi.
4. Menyelenggarakan tata kelola organisasi yang efisien, transparan, akuntabel, bertanggungjawab, dan adil.

PROGRAM STUDI

Fakultas Teknik menyelenggarakan pendidikan jenjang program S-1 untuk Teknik Elektro, Teknik Kimia (regular dan Joint Degree Program), Teknik Industri, Rekayasa Industri, Informatika; Program Profesi Insinyur; dan program S-2 Teknik Kimia (Magister Teknik Kimia).

Informasi terkini mengenai Fakultas Teknik dapat dilihat di :

<https://www.ft.ukwms.ac.id/>

<https://electrical.ukwms.ac.id/>

<https://www.chemeng.ukwms.ac.id/>

<https://www.ie.ukwms.ac.id/>

<https://www.rekayasaindustri.widyamandala.ac.id/>

<https://informatics.ukwms.ac.id/>

E-mail : dean-eng@ukwms.ac.id

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FACULTY OF ENGINEERING**VISION**

The Faculty of Engineering of Widya Mandala Catholic University Surabaya aspires to become an open engineering education center, recognized nationally and internationally in industrial and information technology, contributing to and improving community welfare while guided by Pancasila values and Catholic principles.

MISSION

1. To organize education that produces graduates with integrity, competence, and professionalism in the field of industrial and information technology.
2. To conduct research and community service in the field of industrial and information technology to improve community welfare.
3. To expand pentahelix networking partnerships for the development of the Tri Dharma of Higher Education.
4. To implement efficient, transparent, accountable, responsible, and fair organizational governance.

STUDY PROGRAMS

The Faculty of Engineering offers undergraduate programs (S-1) in Electrical Engineering, Chemical Engineering (regular and Joint Degree Program), Industrial Engineering, Industrial Engineering (PSDKU), Informatics; Professional Engineer Program; and Master's Program in Chemical Engineering (Magister Teknik Kimia).

For the latest information regarding the Faculty of Engineering, please visit:

<https://www.ft.ukwms.ac.id/>

<https://electrical.ukwms.ac.id/>

<https://www.chemeng.ukwms.ac.id/>

<https://www.ie.ukwms.ac.id/>

<https://www.rekayasaindustri.widyamandala.ac.id/>

<https://informatics.ukwms.ac.id/>

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PROGRAM STUDI TEKNIK ELEKTRO

Seiring dengan perkembangan teknologi dewasa ini, ilmu dari Teknik Elektro berperan penting pada perkembangan bidang teknik biomedika, artificial intelligent (AI) dan sistem internet of things (IoT), energi baru terbarukan atau komputer yang sangat menyokong kehidupan masyarakat. Demikian pula halnya dengan bidang industri yang masih terus berkembang pesat di Indonesia dalam hal otomatisasi control industri, yang juga merupakan salah satu bidang ilmu elektro. Ilmu Teknik Elektro akan terus bertumbuh, dan seiring dengan perkembangan teknologi terutama Revolusi Industri 4.0 menuju era society 5.0 bagaimana upaya penggabungan teknologi dan manusia yang berfokus pada integrasi antara teknologi seperti AI, IoT, dan teknologi robotika dalam meningkatkan kualitas hidup masyarakat. Program Studi Teknik Elektro Universitas Katolik Widya Mandala Surabaya (PSTE-UKWMS) berada di bawah Fakultas Teknik Universitas Katolik Widya Mandala Surabaya. Sebelumnya merupakan Lembaga Pendidikan Elektronika (LPE), pendidikan setara D-3, yang berdiri sejak tahun 1973. Pada tahun 1982 berkembang menjadi program S1 Teknik Elektro. Program Studi Teknik Elektro Universitas Katolik Widya Mandala Surabaya (PS-TE) didirikan pada tanggal 8 Juni 1982 oleh Yayasan Widya Mandala Surabaya dengan SK No. 022/Ya/1982; mendapatkan Status Terdaftar berdasarkan SK Menteri Pendidikan Dan Kebudayaan Republik Indonesia No. 070/O/1985 tanggal 18 Februari 1985. Status Diakui diperoleh pada tanggal 11 September 1995 berdasarkan SK Dirjen Dikti Depdikbud No. 391/Dikti/Kep/1995, dan pada tanggal 22 Desember 1998 telah terakreditasi A berdasarkan Keputusan Badan Akreditasi Nasional Perguruan Tinggi Departemen Pendidikan dan Kebudayaan Republik Indonesia Nomor 002/BAN-PT/Ak-II/XII/1998. PS-TE saat ini terakreditasi B sejak tanggal 24 September 2019 dengan SK No. 3569/SK/BAN-PT/Akred/S/IX/2019. Berdasarkan Surat Keputusan Badan Akreditasi Nasional Perguruan Tinggi No. 2886/SK/BAN-PT/AK-ISK/S/V/2022. Program Studi Teknik Elektro (PSTE) memperoleh status Terakreditasi dengan peringkat "Akreditasi Baik Sekali". Berdasarkan Keputusan LAM Teknik No. 0417/SK/LAM Teknik/AS/VIII/2024, pada tanggal 21 Agustus 2024, Program Studi Teknik Elektro memperoleh status Terakreditasi dengan peringkat "Baik Sekali".

TUJUAN PENDIDIKAN

Tujuan pendidikan yang dicapai oleh Program Studi Teknik Elektro adalah:

1. Menyelenggarakan pendidikan tinggi yang bermutu, berbasis kurikulum yang relevan dan mutakhir, untuk membekali mahasiswa dengan pengetahuan, keterampilan, dan sikap profesional yang dibutuhkan dalam menghadapi perkembangan teknologi dan dinamika global.
2. Menghasilkan lulusan yang kompeten, inovatif, dan adaptif dalam bidang Teknik Elektro, yang mampu memberikan **dampak positif** bagi dunia usaha, industri, dan masyarakat di tingkat nasional maupun internasional, serta menjunjung tinggi nilai-nilai Pancasila dan prinsip-prinsip etika Katolik.
3. Mengembangkan penelitian dan inovasi yang **berdampak nyata** dalam menyelesaikan permasalahan masyarakat serta berkontribusi pada pencapaian tujuan pembangunan berkelanjutan.
4. **Membangun dan menguatkan kolaborasi strategis berbasis pendekatan pentahelix** (akademisi, dunia usaha/industri, pemerintah, masyarakat, dan media) guna meningkatkan relevansi, mutu, serta **dampak luas** dari penyelenggaraan pendidikan, penelitian, dan pengabdian kepada masyarakat dalam bidang Teknik Elektro.

SARANA PENDIDIKAN

PS Teknik Elektro saat ini memiliki peminatan di bidang biomedika, artificial intelligent dan sistem internet of things (IoT), energi baru terbarukan atau komputer yang terdapat dalam bentuk matakuliah pilihan bidang minat. Aplikasi disesuaikan pada bidang minat dan kemampuan praktis yang dilakukan di dalam Laboratorium yang ada di PS Teknik Elektro. Untuk itu disediakan sarana-sarana pendukung dalam sejumlah laboratorium yaitu:

1. Laboratorium Elektronika (laboratorium pendidikan): Laboratorium Rangkaian dan Pengukuran Besaran Listrik, Laboratorium Elektronika, Laboratorium Kontrol Industri, Laboratorium Telekomunikasi, Laboratorium Konversi Energi Listrik.
2. Laboratorium Teknik Biomedika & Teknologi untuk Orang Berkebutuhan Khusus: Laboratorium Teknik Biomedika, Laboratorium Pemrosesan Sinyal Digital, Laboratorium Sistem Mikroprosesor.
3. Laboratorium Telekomunikasi Multimedia & Internet of Things: Laboratorium Jaringan Komputer, Laboratorium Komputasi.

ELECTRICAL ENGINEERING STUDY PROGRAM

In line with current technological developments, Electrical Engineering plays a crucial role in the advancement of biomedical engineering, artificial intelligence (AI), Internet of Things (IoT) systems, renewable energy, and computing, which greatly support community life. Similarly, the industrial sector continues to grow rapidly in Indonesia, particularly in industrial control automation, which is also a field within electrical engineering. Electrical Engineering will continue to evolve alongside technological developments, especially during the Industrial Revolution 4.0 towards the Society 5.0 era, which focuses on integrating technology and humanity, emphasizing technologies such as AI, IoT, and robotics to improve the quality of community life.

The Electrical Engineering Study Program of Widya Mandala Catholic University Surabaya (PSTE-UKWMS) is under the Faculty of Engineering. Previously, it was the Electronics Education Institution (LPE), established in 1973 and offering D-3-level education. In 1982, it developed into an S1 Electrical Engineering program. The Electrical Engineering Study Program was established on June 8, 1982, by the Widya Mandala Surabaya Foundation under Decree No. 022/Ya/1982; it obtained Registered Status based on the Minister of Education and Culture of the Republic of Indonesia Decree No. 070/O/1985 dated February 18, 1985. Recognized Status was obtained on September 11, 1995, based on the Director General of Higher Education Decree No. 391/Dikti/Kep/1995, and on December 22, 1998, obtained an "A" accreditation based on the National Accreditation Board for Higher Education Decree No. 002/BAN-PT/Ak-II/XII/1998.

Currently, the study program has been accredited "B" since September 24, 2019, with Decree No. 3569/SK/BAN-PT/Akred/S/IX/2019. Based on the National Accreditation Board for Higher Education Decree No. 2886/SK/BAN-PT/AK-ISK/S/V/2022, the Electrical Engineering Study Program obtained "Very Good" accreditation status. Based on LAM Teknik Decree No. 0417/SK/LAM Teknik/AS/VIII/2024, dated August 21, 2024, the Electrical Engineering Study Program obtained "Very Good" accreditation status.

EDUCATIONAL OBJECTIVES

The educational objectives of the Electrical Engineering Study Program are:

1. To organize high-quality higher education based on a relevant and up-to-date curriculum, equipping students with knowledge, skills, and professional attitudes needed to face technological developments and global dynamics.
2. To produce competent, innovative, and adaptive graduates in the field of Electrical Engineering who can make **positive impacts** on business, industry, and society at national and international levels, while upholding Pancasila values and Catholic ethical principles.
3. To develop research and innovation that have **real impact** in solving community problems and contributing to the achievement of sustainable development goals.
4. **To build and strengthen strategic collaboration based on the pentahelix approach** (academics, business/industry, government, community, and media) to enhance the relevance, quality, and **broad impact** of the implementation of education, research, and community service in the field of Electrical Engineering.

EDUCATIONAL FACILITIES

The Electrical Engineering Study Program currently has specializations in biomedical engineering, artificial intelligence and internet of things (IoT) systems, renewable energy, and computing, offered as elective courses. Applications are tailored to the field of interest and practical skills conducted in the laboratories. Supporting facilities are provided in several laboratories:

1. Electronics Laboratory (educational laboratory): Electrical Circuit and Measurement Laboratory, Electronics Laboratory, Industrial Control Laboratory, Telecommunications Laboratory, Electrical Energy Conversion Laboratory.
2. Biomedical Engineering & Technology for Persons with Special Needs Laboratory: Biomedical Engineering Laboratory, Digital Signal Processing Laboratory, Microprocessor Systems Laboratory
3. Multimedia Telecommunications & Internet of Things Laboratory: Computer Networks Laboratory, Computing Laboratory

4. Laboratorium Instrumentasi & Big Data: Laboratorium Instrumentasi.
5. Laboratorium Otomasi Industri & Robotik: Laboratorium Mekanik dan Robotik.

Semua laboratorium terhubung dengan Jaringan Komputer dan Internet. Ruang kuliah yang dilengkapi dengan AC. Alat penunjang perkuliahan: Multimedia Projector. Perpustakaan yang dilengkapi dengan buku-buku yang up to date.

Untuk menunjang kemajuan akademik, mahasiswa juga mendapat dukungan dalam bentuk:

- Layanan penasihat akademik, tutorial, praktikum, bimbingan Kerja Praktek, serta Skripsi,
- Beasiswa, kegiatan penelitian dan pengabdian kepada masyarakat dosen yang melibatkan mahasiswa.

KEUNGGULAN

1. PSTE-WM termasuk program studi bermutu, terakreditasi dengan peringkat Baik Sekali (2022-2027) dari Badan Akreditasi Nasional Perguruan Tinggi (BAN-PT).
2. Merupakan salah satu dari sedikit Jurusan Teknik Elektro yang mempunyai konsentrasi di bidang elektronika kedokteran (biomedical engineering).
3. Kerja sama dengan universitas luar negeri dalam bentuk pertukaran staf dan penelitian, serta kegiatan mahasiswa seperti Summer Camp, Winter Camp, dan beasiswa untuk melanjutkan ke program magister.
4. Merintis penerapan student centered learning (dengan dukungan hibah dari DIKTI), dengan menerapkan TIK untuk mendukung proses belajar - mengajar.
5. Peran aktif dan antusiasme mahasiswa dalam berbagai kegiatan lomba di tingkat nasional.

SISTEM PENDIDIKAN

PSTE-UKWMS menggunakan sistem Satuan Kredit Semester (SKS) dengan beban studi total 144 SKS, terdistribusi dalam 8 semester (4 tahun). Kurikulum yang ada menunjang pembentukan kompetensi lulusan di bidang teknik elektro yang ditekankan pada Biomedika, Artificial Intelligence dan Sistem IoT, Energi Baru Terbarukan atau komputer

Permendikbud No. 3 Tahun 2020 tentang Standar Nasional Pendidikan Tinggi (SNPT) dan Buku Panduan Merdeka Belajar Kampus Merdeka (Mendikbud, 2024) mensyaratkan semua program studi pada Perguruan Tinggi di seluruh Indonesia untuk memberi hak pada mahasiswa untuk belajar di luar program studi asalnya selama lamanya 3 semester atau setara 20 SKS per semester. Untuk itu, kurikulum PS-TE FT UKWMS dilakukan penyesuaian sehingga ada dua bentuk kurikulum yang ditawarkan. Pertama yaitu kurikulum reguler (Kurikulum 8:0) yang artinya 8 semester ditempuh sesuai kurikulum PS-TE UKWMS 2025. Kedua yaitu kurikulum Merdeka Belajar Kampus Merdeka (MBKM) yang terdiri dari 3 skema yaitu Kurikulum 7:1, Kurikulum 6:2, dan Kurikulum 5:3 yang memberikan kesempatan pada mahasiswa untuk belajar di luar PS-TE UKWMS setara beban belajar 1 semester, 2 semester, atau 3 semester. Kurikulum MBKM mulai semester 5 hingga semester 8. Bentuk pembelajaran di luar PS yang disediakan dalam kurikulum ini ada 4 bentuk dari 8 bentuk kegiatan MBKM yang ditawarkan DIKTI (DIKTI, 2024) yaitu pertukaran mahasiswa, magang/praktik industri, kuliah kerja nyata tematik (KKNT)/membangun desa, dan penelitian. Bentuk kegiatan pembelajaran pertukaran mahasiswa dapat berupa kuliah di PS lain di UKWMS, PS-TE di Perguruan Tinggi (PT) lain, dan PS lain di PT lain. Mahasiswa mempunyai hak untuk memilih skema kurikulum dan dengan persetujuan dosen Penasihat Akademik yang dilakukan saat memprogram KRS semester berikutnya setelah lulus minimum 75 SKS.

PROSPEK KERJA LULUSAN

Sarjana Teknik Elektro dapat bekerja hampir di bidang yang melibatkan biomedika, teknologi elektronika, teknologi informasi dan komunikasi, kontrol otomatis, dan kelistrikan. Beberapa bidang industri atau bidang jasa dimana alumni PSTE-UKWMS berkarya yaitu: industri elektronika, teknologi informasi, otomasi industri, peralatan laboratorium kedokteran, otomotif, modifikasi alutsista, BUMN Perbankan, pendidikan. Sejumlah lulusan juga mendirikan dan menjalankan usaha sendiri. Beberapa tempat alumni bekerja:

1. Intel Indonesia Corporation
2. PT. Hartono Istana Teknologi (Polytron)
3. PT. PLN
4. PT. Telkom
5. PT. Sinar Baja Electric

4. Instrumentation & Big Data Laboratory: Instrumentation Laboratory
5. Industrial Automation & Robotics Laboratory: Mechanics and Robotics Laboratory

All laboratories are connected to Computer Networks and the Internet. Lecture rooms are equipped with air conditioning. Teaching support equipment includes Multimedia Projectors. The library is equipped with up-to-date books.

To support academic progress, students also receive support in the form of:

- Academic advisory services, tutorials, practicums, internship guidance, and thesis guidance
- Scholarships, research activities, and community service involving students

ADVANTAGES

1. PSTE-WM is a quality study program with "Very Good" accreditation (2022-2027) from the National Accreditation Board for Higher Education (BAN-PT).
2. One of the few Electrical Engineering Departments in Indonesia with a concentration in biomedical electronics (biomedical engineering).
3. Collaboration with foreign universities in staff exchange, research, and student activities such as Summer Camp, Winter Camp, and scholarships for master's degree continuation.
4. Pioneering the implementation of student-centered learning (supported by DIKTI grants), applying ICT to support the teaching-learning process.
5. Active role and enthusiasm of students in various national-level competitions.

EDUCATION SYSTEM

PSTE-UKWMS uses the Semester Credit Unit system (SKS) with a total study load of 144 SKS, distributed over 8 semesters (4 years). The curriculum supports the development of graduate competencies in electrical engineering with emphasis on Biomedicine, Artificial Intelligence and IoT Systems, and Renewable Energy.

Permendikbud No. 3 of 2020 concerning National Standards for Higher Education (SNPT) and the Independent Learning - Independent Campus Guidebook (Mendikbud, 2024) require all study programs in Higher Education throughout Indonesia to grant students the right to study outside their original study program for up to 3 semesters or equivalent to 20 SKS per semester. Therefore, the curriculum of PS-TE FT UKWMS has been adjusted to offer two forms of curriculum. First, the regular curriculum (Curriculum 8:0), meaning 8 semesters taken in accordance with the PS-TE UKWMS 2025 curriculum. Second, the Independent Learning - Independent Campus (MBKM) curriculum consists of 3 schemes: Curriculum 7:1, Curriculum 6:2, and Curriculum 5:3, providing opportunities for students to study outside PS-TE UKWMS, equivalent to 1, 2, or 3 semesters of study load. The MBKM curriculum runs from semesters 5 to 8. Four forms of learning outside the study program are offered from the 8 MBKM activities offered by DIKTI (DIKTI, 2024): student exchange, internships/industrial practice, thematic community service (KKNT)/village development, and research. Student exchange activities can take the form of lectures in other study programs at UKWMS, PS-TE at other Higher Education Institutions, or other study programs at other Higher Education Institutions. Students have the right to choose a curriculum scheme with the approval of their Academic Advisor, conducted when programming the KRS for the next semester after passing a minimum of 75 SKS.

GRADUATE CAREER PROSPECTS

Electrical Engineering graduates can work in almost any field involving biomedicine, electronics technology, information and communication technology, automatic control, and electrical systems. Some industrial or service sectors where PSTE-UKWMS alumni work include: electronics industry, information technology, industrial automation, medical laboratory equipment, automotive, defense equipment modification, state-owned banking enterprises, and education. Several graduates have also established and run their own businesses.

Some companies where alumni work:

1. Intel Indonesia Corporation
2. PT. Hartono Istana Teknologi (Polytron)
3. PT. PLN
4. PT. Telkom
5. PT. Sinar Baja Electric

6. PT. Indoyoke
7. CV. Innovative Electronics
8. PT. Sarana Reka Eltra Kencana (Alat Kesehatan)
9. PT. Tata Solusi Pratama (Industri HVAC)
10. PT. VektordayaMekatronika (Uninterruptible Power Supply)
11. PT. Star Cosmos, PT. Berca Indonesia
12. PT. Jaya Indah EnggalMandiri (Navigation, Communication, and Weapon Control System)
13. Lasika Production (Sound System, Audio Visual), CV. Smart Home Solusindo
14. PT. Medika Karya Sentosa (Pemrosesan Natural Gas menjadi LPG)
15. Sekolah Tinggi Ilmu Komputer (STIKOM) Surabaya
16. University of California –San Diego –USA
17. Digital Studio Workshop – School of Computer Graphics
18. PT. Arta Boga Cemerlang (Group Orang Tua)

Beberapa bidang wirausaha yang dijalankan alumni: industri kecil pembuatan transformator, industri kecil pembuatan AVR (Automatic Voltage Regulator), instalasi sistem otomasi industri, instalasi security and surveillance system, distributor-retailer komponen elektronik, perawatan dan retailer komputer, jasa kesehatan lingkungan, international packing-moving-and-transportation, industri makanan.

KERJA SAMA

Program Studi Teknik Elektro telah menjalin hubungan kerja sama aktif dengan beberapa perguruan tinggi, antara lain:

1. Taiwan Tech, Taiwan
2. National Chung Cheng University, Taiwan
3. Shibaura Institute of Technology, Jepang
4. Osaka Institute of Technology, Jepang

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6. PT. Indoyoke
7. CV. Innovative Electronics
8. PT. Sarana Reka EltraKencana (Medical Equipment)
9. PT. Tata Solusi Pratama (HVAC Industry)
10. PT. VektordayaMekatronika (Uninterruptible Power Supply)
11. PT. Star Cosmos, PT. Berca Indonesia
12. PT. Jaya Indah EnggalMandiri (Navigation, Communication, and Weapon Control System)
13. Lasika Production (Sound System, Audio Visual), CV. Smart Home Solusindo
14. PT. Medika Karya Sentosa (Natural Gas Processing to LPG)
15. Surabaya College of Computer Science (STIKOM)
16. University of California - San Diego - USA
17. Digital Studio Workshop - School of Computer Graphics
18. PT. Arta Boga Cemerlang (Orang Tua Group)

Entrepreneurial fields pursued by alumni: small transformer manufacturing, small AVR (Automatic Voltage Regulator) manufacturing, industrial automation system installation, security and surveillance system installation, electronic component distributor-retailer, computer maintenance and retail, environmental health services, international packing-moving-and-transportation, food industry.

PARTNERSHIPS

The Electrical Engineering Study Program has established active cooperation with several universities, including:

1. Taiwan Tech, Taiwan
2. National Chung Cheng University, Taiwan
3. Shibaura Institute of Technology, Japan
4. Osaka Institute of Technology, Japan

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PROGRAM STUDI TEKNIK KIMIA

Program Studi Teknik Kimia UKWMS berkomitmen untuk terus meningkatkan mutu pendidikan secara profesional, adaptif, dan berdampak. Kurikulum disusun sebagai respons terhadap dinamika perkembangan ilmu pengetahuan, teknologi, kebutuhan industri, serta tantangan sosial dan lingkungan. Penyusunan kurikulum mengacu pada standar nasional dan internasional, termasuk KKNI, kebijakan Kemdiktisaintek, IABEE, dan panduan dari APTEKIM, serta selaras dengan visi, misi, nilai, dan tujuan institusi serta program studi. Kurikulum 2025 dirancang berbasis Outcome-Based Education (OBE) dan mengintegrasikan prinsip Sustainable Development Goals (SDGs), serta mendukung penuh inisiatif Kampus Berdampak sebagai bentuk implementasi pembelajaran yang relevan, kontekstual, dan bermanfaat bagi masyarakat. Dengan struktur kurikulum yang fleksibel, peminatan yang terarah pada pencapaian CPL dan PPM, serta beragam program unggulan seperti double degree, RPL, dan program internasional, Teknik Kimia UKWMS berupaya mencetak lulusan yang kompeten, inovatif, dan siap berkontribusi nyata di tingkat lokal maupun global.

Profil Profesional Mandiri (PPM)

Program Studi Teknik Kimia UKWMS mencetak lulusan dengan kriteria:

1. **Chemical Engineer yang Inovatif dan Berwawasan Lingkungan** - Mampu merancang, mengimplementasikan, dan mengoptimalkan proses teknik kimia secara efisien dan berkelanjutan, serta memiliki kesadaran tinggi terhadap isu lingkungan dan tanggung jawab sosial.
2. **Technopreneur di Bidang Teknologi Kimia** - Mampu mengidentifikasi peluang berbasis teknologi kimia, merancang solusi inovatif, serta membangun dan mengelola usaha berbasis sains dan kearifan lokal secara beretika dan berdaya saing.
3. **Peneliti dan Akademisi di Bidang Teknik Kimia** - Mampu mengembangkan pengetahuan dan teknologi baru melalui riset yang relevan, serta menyebarkan hasilnya melalui publikasi ilmiah dan kegiatan pendidikan tinggi.

Capaian Pembelajaran Lulusan

1. **Problem Solving** - Mampu menerapkan ilmu dasar teknik dan prinsip teknik kimia dalam mengidentifikasi, merumuskan, dan menyelesaikan permasalahan teknik kimia yang kompleks secara sistematis dan komprehensif.
2. **Perancangan Proses dan Sistem** - Mampu merancang proses, sistem, dan alat proses teknik kimia secara realistis dengan mempertimbangkan faktor ekonomi, kesehatan, keselamatan publik, sosial, lingkungan, dan keberlanjutan.
3. **Riset dan Inovasi** - Mampu merancang dan melaksanakan eksperimen di laboratorium atau lapangan, serta menganalisis dan menginterpretasikan data untuk menghasilkan kesimpulan rekayasa yang valid.
4. **Komunikasi dan Etika Profesi** - Mampu berkomunikasi secara efektif secara lisan dan tulisan, serta menjunjung tinggi tanggung jawab etis, profesi, dan norma yang berlaku dalam praktik keteknikan.
5. **Kerjasama dan Manajerial** - Mampu merencanakan, melaksanakan, dan mengevaluasi proyek teknik kimia secara kolaboratif dalam tim multidisiplin dan multikultural dalam jangka waktu tertentu.
6. **Technopreneurship** - Mampu memahami dan menerapkan prinsip technopreneurship, serta memanfaatkan potensi lokal untuk merancang solusi inovatif yang menjawab kebutuhan masyarakat.
7. **Pengembangan Diri** - Mampu menyadari pentingnya peningkatan kompetensi diri secara berkelanjutan, serta memanfaatkan teknologi informasi dalam pembelajaran dan pengambilan keputusan teknik.

TUJUAN PENDIDIKAN

1. Menjadi pelopor dalam pengembangan inovasi dan riset terapan di bidang teknik kimia, khususnya dalam teknologi proses, energi terbarukan, dan material maju berbasis keberlanjutan.
2. Berkembang secara profesional dengan semangat technopreneurship dan kepemimpinan, serta mampu menciptakan nilai tambah melalui inovasi teknologi dan pemanfaatan kearifan lokal.

CHEMICAL ENGINEERING STUDY PROGRAM

The UKWMS Chemical Engineering Study Program is committed to continuously improving the quality of education professionally, adaptively, and with impact. The curriculum is designed as a response to the dynamics of scientific development, technology, industrial needs, and social and environmental challenges. Curriculum development involves aligning national and international standards, including KKNI, Kemdiktisaintek policies, IABEE, and APTEKIM guidelines, with the institution's vision, mission, values, and objectives, as well as the study program's. The 2025 Curriculum is designed on the principles of Outcome-Based Education (OBE), integrates the principles of the Sustainable Development Goals (SDGs), and fully supports the Impactful Campus initiative as a means of implementing relevant, contextual, and beneficial learning for the community. With a flexible curriculum structure, specializations focused on achieving CPL and PPM, and various flagship programs such as double degree, RPL, and international programs, Chemical Engineering UKWMS strives to produce competent, innovative graduates ready to make real contributions at the local and global levels.

Independent Professional Profile (PPM)

The UKWMS Chemical Engineering Study Program produces graduates with the following criteria:

- 1. Innovative and Environmentally Conscious Chemical Engineer** - Able to design, implement, and optimize chemical engineering processes efficiently and sustainably, with high awareness of environmental issues and social responsibility.
- 2. Technopreneur in Chemical Technology** - Able to identify technology-based opportunities in chemistry, design innovative solutions, and build and manage science-based businesses with local wisdom ethically and competitively.
- 3. Researcher and Academician in Chemical Engineering** - Able to develop new knowledge and technology through relevant research, and disseminate results through scientific publications and higher education activities.

Graduate Learning Outcomes

- 1. Problem Solving** - Able to apply basic engineering sciences and chemical engineering principles to identify, formulate, and solve complex chemical engineering problems systematically and comprehensively.
- 2. Process and System Design** - Able to design chemical processes, systems, and process equipment realistically considering economic factors, health, public safety, social, environmental, and sustainability aspects.
- 3. Research and Innovation** - Able to design and conduct laboratory or field experiments, analyze and interpret data to produce valid engineering conclusions.
- 4. Communication and Professional Ethics** - Able to communicate effectively orally and in writing, and uphold ethical, professional responsibilities, and applicable norms in engineering practice.
- 5. Collaboration and Managerial Skills** - Able to plan, implement, and evaluate chemical engineering projects collaboratively in multidisciplinary and multicultural teams within a specified timeframe.
- 6. Technopreneurship** - Able to understand and apply technopreneurship principles and utilize local potential to design innovative solutions that address community needs.
- 7. Self-Development** - Able to recognize the importance of continuous self-competency improvement and utilize information technology in learning and engineering decision-making.

EDUCATIONAL OBJECTIVES

1. To become a pioneer in developing innovation and applied research in chemical engineering, particularly in process technology, renewable energy, and advanced sustainability-based materials.
2. To develop professionally with technopreneurship spirit and leadership, capable of creating added value through technological innovation and utilization of local wisdom.

3. Melanjutkan pendidikan ke jenjang yang lebih tinggi dan berkomitmen pada pengembangan diri sepanjang hayat, termasuk melalui sertifikasi profesional, pelatihan teknis, maupun kontribusi ilmiah.
4. Berperan aktif dalam pembangunan berkelanjutan dan penguatan tanggung jawab sosial, baik di lingkungan kerja, komunitas, maupun melalui pengambilan kebijakan yang beretika.
5. Menerapkan kompetensi teknis dan soft skills secara intergratif, termasuk kemampuan analitis, komunikasi, kolaborasi multidisiplin, dan adaptasi terhadap perkembangan teknologi industri global.

SISTEM PENDIDIKAN

Dengan semangat Kurikulum Berdampak, Kurikulum Teknik Kimia dirancang untuk menghasilkan lulusan yang kompeten, inovatif, berwawasan lingkungan, dan siap bersaing secara global.

Program Reguler

Program Reguler dengan gelar Sarjana Teknik (S.T.) di Teknik Kimia memiliki 144 SKS yang dapat ditempuh dalam 8 semester:

Semester	Ciri Pembelajaran
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1–2	Fondasi sains dan pengembangan karakter 3–4
3–4	Dasar teknik kimia dan keterampilan praktis
5–6	Aplikasi teknik dan pemecahan masalah melalui Studi Kasus
7–8	Integrasi dan penyelesaian studi (Meliputi: Skripsi, Kerja Pratek)

Bidang Peminatan:

Dalam Kurikulum 2025, peminatan mahasiswa diarahkan secara strategis untuk mendukung pencapaian Capaian Pembelajaran Lulusan (CPL) dan Profil Profesional Mandiri (PPM). Program Studi Teknik Kimia UKWMS menawarkan tiga kelompok peminatan utama yang merepresentasikan fokus kekuatan program studi dan kebutuhan masa depan industri:

1. Green and Sustainable Technology (GST): Fokus pada teknologi proses ramah lingkungan, energi terbarukan, pengolahan limbah, dan keberlanjutan industri kimia.
2. Biotechnology and Bioengineering (BB): Menekankan pengembangan proses bioproses, rekayasa mikroorganisme, dan aplikasi teknik kimia dalam farmasi, pangan, dan bioteknologi industri.
3. Material and Process Engineering (MPE): Mencakup rekayasa material fungsional, pengembangan proses pemisahan dan sintesis material, serta teknologi proses lanjutan.

Program Internasional

Program Internasional dalam bentuk Gelar Bersama (Joint Degree Program, JDP) memberikan kesempatan kepada mahasiswa untuk memperoleh pengalaman akademik global melalui kerja sama dengan perguruan tinggi luar negeri. Mahasiswa yang mengikuti program ini dapat menghabiskan sebagian masa studi di luar negeri dengan pengakuan kredit dan gelar ganda Sarjana Teknik (S.T.) dan Bachelor of Science (B.Sc.).

Mitra internasional PSTK untuk Joint Degree Program meliputi:

- National Taiwan University of Science and Technology (NTUST), Taiwan
- Beijing University of Chemical Technology (BUCT), China

Program Rekognisi Pembelajaran Lampau (RPL)

Program RPL ditujukan bagi individu yang memiliki pengalaman kerja relevan atau telah mengikuti pendidikan non-formal, informal, atau pelatihan industri. Berdasarkan penilaian portofolio, pengalaman tersebut dapat dikonversi menjadi SKS yang diakui dalam kurikulum akademik. Program Studi Teknik Kimia UKWMS telah resmi menyelenggarakan RPL sejak tahun 2023 dengan persetujuan Ditjen Dikti Ristek. RPL memberikan jalur alternatif dan percepatan studi bagi peserta didik berpengalaman.

3. To pursue higher education and commit to lifelong self-development, including professional certification, technical training, and scientific contributions.
4. To actively participate in sustainable development and strengthen social responsibility, both in the work environment, communities, and through ethical policymaking.
5. To apply technical competencies and soft skills integratively, including analytical abilities, communication, multidisciplinary collaboration, and adaptation to global industrial technology developments.

EDUCATION SYSTEM

With the spirit of the Impactful Curriculum, the Chemical Engineering Curriculum is designed to produce competent, innovative, environmentally conscious graduates ready to compete globally.

Regular Program

The Regular Program with a Bachelor of Engineering (S.T.) degree in Chemical Engineering has 144 SKS which can be completed in 8 semesters:

Semester	Learning Characteristics
1-2	Science foundations and character development
3-4	Basic chemical engineering and practical skills
5-6	Engineering application and problem solving through Case Studies
7-8	Integration and study completion (Including: Thesis, Internship)

Specialization Fields:

In the 2025 Curriculum, student specializations are strategically directed to support the achievement of Graduate Learning Outcomes (CPL) and Independent Professional Profiles (PPM). The UKWMS Chemical Engineering Study Program offers three main specialization groups representing the program's strengths and future industrial needs:

1. Green and Sustainable Technology (GST): Focuses on environmentally friendly process technology, renewable energy, waste processing, and chemical industry sustainability.
2. Biotechnology and Bioengineering (BB): Emphasizes bioprocess development, microorganism engineering, and chemical engineering applications in pharmaceuticals, food, and industrial biotechnology.
3. Material and Process Engineering (MPE): Covers functional material engineering, development of separation processes and material synthesis, and advanced process technology.

International Program

The International Program in the form of a Joint Degree Program (JDP) provides opportunities for students to gain global academic experience through collaboration with foreign universities. Students participating in this program can spend part of their study period abroad with credit recognition and dual degrees: Bachelor of Engineering (S.T.) and Bachelor of Science (B.Sc.).

International partners for the Joint Degree Program include:

- National Taiwan University of Science and Technology (NTUST), Taiwan
- Beijing University of Chemical Technology (BUCT), China

Recognition of Prior Learning (RPL) Program

The RPL program is intended for individuals who have relevant work experience or have completed non-formal, informal education, or industrial training. Based on portfolio assessment, this experience can be converted into SKS recognized in the academic curriculum. The UKWMS Chemical Engineering Study Program has officially implemented RPL since 2023 with approval from the Directorate General of Higher Education. RPL provides alternative pathways and study acceleration for experienced learners.

Program Double Degree

Sejalan dengan kebutuhan lintas disiplin dan era transformasi digital, Program Studi Teknik Kimia UKWMS menawarkan skema Double Degree Teknik Kimia – Informatika. Program ini memungkinkan mahasiswa memperoleh kompetensi ganda dalam bidang teknik kimia dan ilmu komputer seperti data sains, automasi industri, dan pemodelan sistem. Kurikulum disusun agar fleksibel, dengan sebagian besar perkuliahan disesuaikan melalui jalur pilihan dan MBKM. Mahasiswa akan mendapatkan dua gelar sarjana setelah memenuhi seluruh persyaratan masing-masing program studi dengan masa studi minimal 4,5 hingga 5 tahun, yaitu gelar Sarjana Teknik (S.T.) dari Program Studi Teknik Kimia dan Sarjana Komputer (S. Kom.).

KEUNGGULAN

1. Fokus pada Green Sustainable Technology dan Inovasi. - Program Studi Teknik Kimia UKWMS menekankan pada pengembangan teknologi hijau dan energi terbarukan, yang sejalan dengan tren global dan kebutuhan industri.
2. Kolaborasi dengan Industri dan Lembaga Riset. - Kurikulum dan penelitian di Program Studi Teknik Kimia UKWMS didukung oleh kolaborasi dengan industri dan lembaga riset terkemuka, memastikan lulusan memiliki keterampilan yang relevan dengan kebutuhan pasar.
3. Prestasi dalam Penelitian dan Publikasi. - Dengan pencapaian Ranking SINTA peringkat 1 selama 3 tahun berturut-turut dan penghargaan bergengsi seperti Habibie Awards, Program Studi Teknik Kimia UKWMS menunjukkan keunggulan dalam penelitian dan inovasi.
4. Pengembangan Soft Skills dan Technopreneurship. -Program Studi Teknik Kimia UKWMS tidak hanya fokus pada hard skills, tetapi juga mengembangkan soft skills dan jiwa kewirausahaan, mempersiapkan lulusan untuk menjadi pemimpin dan inovator di bidangnya.

FASILITAS

Laboratorium dan Fasilitas Praktis:

- Laboratorium Kimia Analisa (KA)
- Laboratorium Kimia Organik dan Kimia Fisika (KO/KF)
- Laboratorium Teknologi Bioproses
- Laboratorium Operasi Teknik Kimia
- Laboratorium Teknologi Material
- Laboratorium Teknologi Proses
- Pusat Instrumentasi: Analisa Instrumen
- Workshop: Sistem pengolahan limbah Fenton
- Pusat Kolaborasi: Nir Limbah Hijau dan Berkelanjutan

KERJA SAMA

1. Badan Riset dan Inovasi Nasional (BRIN)
2. National Taiwan University of Science and Technology, Taiwan
3. Beijing University of Chemical Technology, China
4. Zhejiang University of Technology, China
5. Institut Teknologi Sepuluh Nopember, Surabaya
6. Cantho University, Vietnam
7. Osaka Institute Technology (OIT), Jepang
8. Shibaura Institute of Technology, Jepang
9. PT. Riau Andalan Pulp dan Paper
10. 1Orang Tua Group
11. PT. UBS
12. dan lain-lain

PROSPEK KERJA

Bidang Profesi Sarjana Teknik Kimia:

1. Riset dan Inovasi
2. Perencanaan Proses Industri
3. Perancangan Alat Industri
4. Perancangan Sistem Pengendalian Lingkungan

Double Degree Program

In line with cross-disciplinary needs and the digital transformation era, the UKWMS Chemical Engineering Study Program offers a Double Degree scheme in Chemical Engineering - Informatics. This program allows students to gain dual competencies in chemical engineering and computer science such as data science, industrial automation, and system modeling. The curriculum is designed flexibly, with most courses adjusted through elective and MBKM pathways. Students will receive two bachelor's degrees after fulfilling all requirements of each study program with a minimum study period of 4.5 to 5 years: Bachelor of Engineering (S.T.) from Chemical Engineering and Bachelor of Computer Science (S.Kom.).

ADVANTAGES

1. Focus on Green Sustainable Technology and Innovation - The UKWMS Chemical Engineering Study Program emphasizes the development of green technology and renewable energy, aligned with global trends and industrial needs.
2. Collaboration with Industry and Research Institutions - The curriculum and research are supported by collaboration with leading industries and research institutions, ensuring graduates have skills relevant to market needs.
3. Excellence in Research and Publication - With SINTA Ranking rank 1 for 3 consecutive years and prestigious awards such as the Habibie Awards, the UKWMS Chemical Engineering Study Program demonstrates excellence in research and innovation.
4. Soft Skills and Technopreneurship Development - The program focuses not only on hard skills but also develops soft skills and entrepreneurial spirit, preparing graduates to become leaders and innovators in their field.

FACILITIES

Laboratories and Practical Facilities:

- Analytical Chemistry Laboratory (KA)
- Organic Chemistry and Physical Chemistry Laboratory (KO/KF)
- Bioprocess Technology Laboratory
- Chemical Engineering Operations Laboratory
- Material Technology Laboratory
- Process Technology Laboratory
- Instrumentation Center: Instrument Analysis
- Workshop: Fenton waste processing system
- Collaboration Center: Green and Sustainable Zero Waste

PARTNERSHIPS

1. National Research and Innovation Agency (BRIN)
2. National Taiwan University of Science and Technology, Taiwan
3. Beijing University of Chemical Technology, China
4. Zhejiang University of Technology, China
5. SepuluhNopember Institute of Technology, Surabaya
6. Cantho University, Vietnam
7. Osaka Institute Technology (OIT), Japan
8. Shibaura Institute of Technology, Japan
9. PT. Riau Andalan Pulp and Paper
10. Orang Tua Group
11. PT. UBS
12. and others

CAREER PROSPECTS

Professional Fields for Chemical Engineering Graduates:

1. Research and Innovation
2. Industrial Process Planning
3. Industrial Equipment Design
4. Environmental Control System Design

5. Pengoperasian Pabrik
6. Pengendalian Mutu / Quality Control
7. Konsultan Teknik & Manajemen Industri
8. Akademisi
9. Sales Engineer
10. Penganalisa kredit dalam dunia perbankan
11. Wiraswata

Email : hod-chemeng@ukwms.ac.id

Website: <https://www.chemeng.ukwms.ac.id>

5. Plant Operations
6. Quality Control
7. Engineering & Industrial Management Consultant
8. Academician
9. Sales Engineer
10. Credit Analyst in Banking
11. Entrepreneur

Email: hod-chemeng@ukwms.ac.id

Website: <https://www.chemeng.ukwms.ac.id>

PROGRAM STUDI TEKNIK INDUSTRI (KAMPUS SURABAYA) DAN REKAYASA INDUSTRI (KAMPUS KOTA MADIUN)

DEFINISI TEKNIK INDUSTRI/REKAYASA INDUSTRI

Adalah program studi yang mempelajari bagaimana merancang, mengelola, dan meningkatkan sistem yang terintegrasi antara manusia, teknologi, informasi, material, dan proses bisnis untuk mencapai produktivitas, efisiensi, kualitas, serta keberlanjutan. Mahasiswa Teknik/Rekayasa Industri tidak hanya dibekali kemampuan teknis di bidang perancangan sistem produksi, logistik, ergonomi, kualitas, dan analisis data, tetapi juga keterampilan kepemimpinan, komunikasi, pemecahan masalah, serta pengambilan keputusan berbasis data yang sangat dibutuhkan di era industri modern. Dengan dukungan dosen yang kompeten, fasilitas laboratorium yang memadai, serta berbagai peluang magang dan kerja sama dengan industri, Program Studi Teknik Industri dan Rekayasa Industri UKWMS mempersiapkan mahasiswa menjadi lulusan yang adaptif, inovatif, dan siap berkontribusi di berbagai sektor industri maupun dunia usaha.

TUJUAN PENDIDIKAN

Tujuan Pendidikan Program Studi Teknik Industri UKWMS adalah:

- Menghasilkan lulusan yang berintegritas, kompeten, dan profesional dalam melakukan perancangan dan perbaikan sistem secara terintegrasi dan inovatif dengan menanamkan nilai Pancasila dan prinsip-prinsip Katolik.
- Menghasilkan penelitian dan pengabdian pada masyarakat berbasis keilmuan di bidang teknik industri yang inovatif dan berdampak bagi masyarakat.
- Mengembangkan kolaborasi pentahelix yang sinergis dan berkelanjutan melalui pendidikan, penelitian dan pengabdian kepada masyarakat.

KOMPETENSI LULUSAN

- Mampu mengaplikasikan matematika, sains, TI, dan prinsip teknik untuk memahami dan menganalisis sistem industri.
- Mampu merancang sistem industri dengan mempertimbangkan aspek teknis, ekonomi, sosial, lingkungan, dan keberlanjutan secara partisipatif.
- Mampu melakukan eksperimen serta menganalisis data secara kritis untuk mendukung pengambilan keputusan teknik.
- Mampu mengidentifikasi dan menyelesaikan masalah sistem kompleks dengan solusi yang bertanggung jawab secara teknis dan sosial.
- Mampu menerapkan metode dan teknologi modern dalam pengembangan sistem industri yang efisien dan ramah lingkungan.
- Mampu berkomunikasi efektif secara lisan dan tulisan kepada berbagai pemangku kepentingan lintas disiplin dan budaya.
- Mampu merencanakan dan mengevaluasi tugas teknik dengan memperhatikan sumber daya dan prinsip keberlanjutan.
- Mampu memimpin dan bekerja sama dalam tim multidisiplin serta memberi kontribusi positif bagi organisasi.
- Menunjukkan kepedulian dan etika profesi dalam berkontribusi kepada masyarakat melalui bidang keahliannya.
- Mampu mengelola pembelajaran mandiri dan mengikuti isu-isu terkini untuk pengembangan diri dan profesionalisme.

KEUNGGULAN TEKNIK INDUSTRI

1. Kerjasama internasional yang kuat, sehingga ada kesempatan mengikuti riset, short courses, magang di berbagai negara antara lain, Jepang, Taiwan dan China

INDUSTRIAL ENGINEERING STUDY PROGRAM (SURABAYA CAMPUS) AND INDUSTRIAL ENGINEERING (MADIUN CITY CAMPUS)

DEFINITION OF INDUSTRIAL ENGINEERING

Industrial Engineering is a study program that focuses on designing, managing, and improving integrated systems that connect humans, technology, information, materials, and business processes to achieve productivity, efficiency, quality, and sustainability. Industrial Engineering students are equipped not only with technical skills in production system design, logistics, ergonomics, quality, and data analysis, but also with leadership, communication, problem-solving, and data-driven decision-making skills that are highly needed in the modern industrial era. With competent lecturers, adequate laboratory facilities, and various internship opportunities and industrial partnerships, the Industrial Engineering Study Program prepares students to become adaptive, innovative graduates ready to contribute to various industrial and business sectors.

EDUCATIONAL OBJECTIVES

The educational objectives of the Industrial Engineering Study Program are:

- To produce graduates with integrity, competence, and professionalism in designing and improving integrated and innovative systems by instilling Pancasila values and Catholic principles.
- To produce innovative and impactful research and community service based on industrial engineering knowledge that benefits the community.
- To develop synergistic and sustainable pentahelix collaboration through education, research, and community service.

GRADUATE COMPETENCIES

- Able to apply mathematics, science, IT, and engineering principles to understand and analyze industrial systems.
- Able to design industrial systems considering technical, economic, social, environmental, and sustainability aspects participatively.
- Able to conduct experiments and critically analyze data to support engineering decision-making.
- Able to identify and solve complex system problems with technically and socially responsible solutions.
- Able to apply modern methods and technologies in developing efficient and environmentally friendly industrial systems.
- Able to communicate effectively orally and in writing to various stakeholders across disciplines and cultures.
- Able to plan and evaluate engineering tasks considering resources and sustainability principles.
- Able to lead and collaborate in multidisciplinary teams and contribute positively to organizations.
- Demonstrate professional ethics and concern in contributing to society through their field of expertise.
- Able to manage independent learning and follow current issues for self-development and professionalism.

ADVANTAGES OF INDUSTRIAL ENGINEERING

1. Strong international cooperation, providing opportunities for research, short courses, and internships in various countries including Japan, Taiwan, and China.

2. Kesempatan mengikuti special programs, yaitu:
GATE (Global Academic and Training Exchange) antara lain: Program 3+1+i dengan St. John University Taiwan, program research internship dengan NTUST, Taiwan serta, Short Courses di Taiwan, China dan Jepang. **Program RPL** yang ditujukan bagi individu yang memiliki pengalaman kerja relevan atau telah mengikuti pendidikan formal maupun informal, ataupun pelatihan industri. PSTI UKWMS memberikan jalur alternatif dan percepatan studi bagi peserta didik yang berpengalaman. **Program Double Degree**, yaitu kesempatan mendapatkan 2 (dua) gelar dengan masa studi 5 tahun. Skema double degree 1, yaitu S1 Teknik Industri - S1 Informatika dengan gelar S.T., S.Kom. Skema double degree 2, yaitu S1 Teknik Industri - S2 Magister Akuntansi dengan gelar S.T., M.Ak.
3. Kurikulum berbasis OBE melalui pendekatan Interdisiplin: menggabungkan ilmu teknik, statistika, manajemen, dan teknologi informasi dalam satu kesatuan yang harmonis
4. Rekognisi tingkat nasional melalui mendapatkan hibah nasional, antara lain hibah penelitian dan pengabdian masyarakat, PKKM 2022, KSK 2020, dll
5. Fasilitas laboratorium yang mendukung.
6. Konsentrasi perkuliahan lebih mengedepankan keahlian perancangan dan rekayasa sistem manufaktur.
7. Masa studi dapat lebih cepat dengan adanya semester sisipan yang memungkinkan mahasiswa dapat lulus dalam 3,5 tahun.
8. Silabus pada sejumlah mata kuliah telah disesuaikan dengan standar sertifikasi profesional, sehingga memberikan peluang bagi mahasiswa untuk mengikuti uji sertifikasi kompetensi di bidang terkait, seperti CSCA (Certified Supply Chain Analyst), ISO 9001: 2015 Awareness Quality Management System, dan Enterprise Resource Planning (ERP / SAP).
9. Kegiatan akademik dan non akademik yang variatif, untuk melatih hardskill dan softskill mahasiswa, melalui Kunjungan Industri, keikutsertaan dalam lomba tingkat nasional dan internasional, dll
10. Program Bahasa Inggris yang memungkinkan setiap mahasiswa mempunyai kemampuan bahasa Inggris yang memadai untuk bersaing dalam era Revolusi Industri 4.0.

PROSPEK KERJA

Industri Manufaktur

1. PPIC (Production Planning & Inventory Control)
2. Quality Control
3. Perancangan Tata Letak Pabrik dan Material Handling
4. Analisis Kelayakan Usaha/Pabrik
5. Perancangan dan audit Sistem Kerja
6. Auditor Sistem Kerja, dll.

Industri Jasa

1. Perancangan Sistem Pelayanan
2. Analisis Kelayakan Industri Jasa
3. Perancang Sistem Informasi Manajemen
4. Auditor Sistem Kerja, dll.

Lain – lain

1. Dosen
2. Konsultan Bidang Jasa dan Manufaktur
3. Wirausahawan, dll.

FASILITAS

1. **Laboratorium Jurusan Teknik Industri**
 - Lab Perancangan Sistem Industri
 - Lab Praktik Industri
 - Lab Rekayasa Sistem Kerja dan ergonomi
 - Lab Sistem Cerdas dan Optimasi
 - Studio Inovasi Industri Berkelanjutan

2. Opportunities to participate in special programs:
GATE (Global Academic and Training Exchange) including: 3+1+i Program with St. John University Taiwan, research internship program with NTUST Taiwan, and Short Courses in Taiwan, China, and Japan. **RPL Program** intended for individuals with relevant work experience or formal and informal education or industrial training. PSTI UKWMS provides alternative pathways and study acceleration for experienced learners. **Double Degree Program**, providing opportunities to obtain 2 degrees with a 5-year study period. Double Degree Scheme: 1: S1 Industrial Engineering - S1 Informatics (S.T., S.Kom.). Double Degree Scheme 2: S1 Industrial Engineering - S2 Master of Accounting (S.T., M.Ak.)
3. OBE-based curriculum through an Interdisciplinary approach: combining engineering, statistics, management, and information technology in a harmonious unity.
4. National recognition through obtaining national grants, including research and community service grants, PKKM 2022, KSK 2020, etc.
5. Supportive laboratory facilities.
6. Lecture concentration prioritizes manufacturing system design and engineering expertise.
7. Faster study period with inter-semester programs allowing students to graduate in 3.5 years.
8. Syllabus for several courses has been adjusted to professional certification standards, providing opportunities for students to take competency certification exams in related fields, such as CSCA (Certified Supply Chain Analyst), ISO 9001:2015 Awareness Quality Management System, and ERP/SAP.
9. Varied academic and non-academic activities to train students' hard skills and soft skills through Industrial Visits, participation in national and international competitions, etc.
10. English Language Program enabling every student to have adequate English proficiency to compete in the Industrial Revolution 4.0 era.

CAREER PROSPECTS

Manufacturing Industry:

1. PPIC (Production Planning & Inventory Control)
2. Quality Control
3. Factory Layout and Material Handling Design
4. Business/Factory Feasibility Analysis
5. Work System Design and Audit
6. Work System Auditor

Service Industry:

1. Service System Design
2. Service Industry Feasibility Analyst
3. Management Information System Designer
4. Work System Auditor

Others:

1. Lecturer
2. Service and Manufacturing Consultant
3. Entrepreneur

FACILITIES

1. Industrial Engineering Department Laboratories:

- Industrial System Design Laboratory
- Industrial Practice Laboratory
- Work System and Ergonomics Engineering Laboratory
- Intelligent and Optimization Systems Laboratory
- Sustainable Industrial Innovation Studio

PSDKU Rekayasa Industri

- Lab. Perancangan Sistem Kerja dan Ergonomi
- Lab. Proses Produksi
- Lab. Gambar Teknik

2. Ruang Kuliah

Semua ruang kuliah dan laboratorium dilengkapi dengan LCD Projector dan Air Conditioner (AC).

3. Jaringan Komputer dan Internet

Akses internet di tiap ruang kelas, laboratorium dan perpustakaan dan disertai WiFi.

4. Perpustakaan

- a) Perpustakaan di Kampus Kalijudan dan Kampus Dinoyo dengan koleksi buku yang lengkap dan up-to-date
- b) Ruang baca yang berisi Textbook dan koleksi skripsi.

SISTEM PENDIDIKAN

Prodi Teknik Industri sudah melaksanakan Kurikulum OBE (Outcome Based Education). Total beban studi meliputi 144 sks dalam 8 semester (4 tahun), yang mencakup:

1. Perkuliahan
2. Praktikum
3. Pertukaran mahasiswa
4. Kerja Praktek
5. Kunjungan ke industri
6. Magang Industri
7. Magang Penelitian
8. Penelitian

KERJA SAMA

1. PT Sinar Baja Elektrik
2. PT Sarimas Permai
3. Riau Andalan Pulp and Paper (RAPP)
4. Edugate Indonesia
5. PT INKA Madiun
6. PT INKA Multi Solusi Service (IMSS) Madiun
7. National Taiwan University of Science and Technology
8. National Chung Cheng University, Taiwan
9. Shibaura Institute of Technology, Jepang
10. Osaka Institute of Technology, Jepang
11. Can Tho University, Vietnam (Staff & Student Exchange)
12. Universitas Katolik Indonesia Atmajaya Jakarta
13. St. John University, Taiwan

Email dan website:

hod-indeng@ukwms.ac.id (PS Teknik Industri)

<https://www.ie.ukwms.ac.id>

hod-osc-indeng@ukwms.ac.id (PSDKU Rekayasa Industri)

PSDKU Industrial Engineering:

- Work System and Ergonomics Design Laboratory
- Production Process Laboratory
- Engineering Drawing Laboratory

2. Lecture Rooms:

All lecture rooms and laboratories are equipped with LCD Projectors and Air Conditioners.

3. Computer Network and Internet:

Internet access in each classroom, laboratory, and library with WiFi.

4. Library:

- a) Library at Kalijudan Campus and Dinoyo Campus with complete and up-to-date book collections.
- b) Reading room containing Textbooks and thesis collections.

EDUCATION SYSTEM

The Industrial Engineering Study Program has implemented the OBE (Outcome Based Education) Curriculum. Total study load includes 144 SKS in 8 semesters (4 years), covering:

1. Lectures
2. Practicums
3. Student Exchange
4. Internship
5. Industrial Visits
6. Industrial Internship
7. Research Internship
8. Research

PARTNERSHIPS

1. PT Sinar Baja Elektrik
2. PT SarimasPermai
3. Riau Andalan Pulp and Paper (RAPP)
4. Edugate Indonesia
5. PT INKA Madiun
6. PT INKA Multi Solusi Service (IMSS) Madiun
7. National Taiwan University of Science and Technology
8. National Chung Cheng University, Taiwan
9. Shibaura Institute of Technology, Japan
10. Osaka Institute of Technology, Japan
11. Can Tho University, Vietnam (Staff & Student Exchange)
12. Atmajaya Catholic University of Indonesia Jakarta
13. St. John University, Taiwan

Email and Website:

hod-indeng@ukwms.ac.id (Industrial Engineering)

<https://www.ie.ukwms.ac.id>

hod-osc-indeng@ukwms.ac.id (PSDKU Industrial Engineering)

PROGRAM STUDI INFORMATIKA

Teknologi saat ini mengalami perkembangan pesat. Perkembangan pesat tersebut dapat membantu perusahaan-perusahaan untuk dapat lebih efisien dan efektif dalam memenangkan persaingan dengan menggunakan bantuan teknologi informasi. Pada era revolusi industri 4.0, data juga mengalami peningkatan sangat pesat dan saat ini disebut dengan big data. Big data memiliki ciri-ciri seperti volume yang sangat besar, pertumbuhan data yang sangat cepat, bentuk data yang beraneka ragam. Perusahaan juga dapat memenangkan persaingan dengan memanfaatkan big data tersebut.

Salah satu profesi penting yang memiliki hubungan yang sangat erat pada perkembangan teknologi dan data adalah profesi informatika. Pada perkembangan teknologi dan data tersebut dibutuhkan kombinasi pengetahuan dan keterampilan untuk membuat perangkat lunak yang dapat membuat proses bisnis lebih efisien dan efektif sehingga dapat memenangkan persaingan. Selain itu juga dibutuhkan kombinasi pengetahuan dan keterampilan untuk dapat menganalisis big data yang ada atau dimiliki oleh perusahaan sehingga dapat memenangkan persaingan. Untuk dapat memenuhi kebutuhan tersebut, sangat diperlukan lulusan program informatika dengan kemampuan analisis dan sintesis yang kuat. PS Informatika UKWMS berdiri berdasarkan Surat Keputusan Menteri Pendidikan dan Kebudayaan Republik Indonesia No. 63/E/O/2024 tanggal 10 Januari 2024 sudah terakreditasi berdasarkan surat dari LAM INFOKOM no.265/SK/LAM-INFOKOM/Ak.S/S/VI/2026.

PROFIL PROFESIONAL MANDIRI

1. Mampu menganalisis persoalan komputasi serta menerapkan prinsip-prinsip komputing dan disiplin ilmu terkait untuk mendesain, mengimplementasi, dan mengevaluasi solusi berbasis komputasi yang sesuai dengan kebutuhan pengguna dan organisasi.
2. Mampu memanfaatkan tren dan peluang dalam perkembangan teknologi informatika atau ilmu komputer untuk mendukung inovasi dan peningkatan kinerja organisasi maupun masyarakat.
3. Mampu berpikir logis, kritis, dan sistematis dalam memanfaatkan pengetahuan di bidang informatika untuk menyelesaikan masalah nyata secara efektif dan efisien.
4. Mampu menunjukkan kinerja mandiri yang bermutu dan terukur, bekerja secara toleran dan bertanggung jawab dalam tim multidisiplin, serta memiliki komitmen terhadap pengembangan diri melalui pembelajaran sepanjang hayat.

CAPAIAN PEMBELAJARAN LULUSAN

1. Mampu menerapkan konsep teoretis ilmu komputer meliputi algoritma, struktur data, matematika komputasi, sistem komputer, dan jaringan sebagai dasar analisis persoalan komputasi.
2. Mampu menerapkan prinsip rekayasa perangkat lunak, analisis kebutuhan, perancangan sistem, serta metode implementasi dan evaluasi solusi komputasi.
3. Mampu menerapkan konsep teknologi terkini di bidang informatika seperti analitik data, kecerdasan buatan, dan keamanan informasi untuk mendukung inovasi.
4. Mampu berpikir logis, kritis, dan sistematis dalam mengidentifikasi, merumuskan, dan menyelesaikan masalah nyata berbasis komputasi secara efektif dan efisien.
5. Mampu mengambil keputusan berbasis analisis data dan informasi serta mengomunikasikan solusi teknis secara jelas dan terukur.
6. Mampu menganalisis kebutuhan pengguna dan organisasi, serta mendesain dan mengimplementasikan solusi berbasis komputasi sesuai standar rekayasa perangkat lunak.
7. Mampu mengevaluasi, memvalidasi, dan meningkatkan kinerja solusi komputasi dengan memanfaatkan perkembangan teknologi terkini untuk mendukung inovasi organisasi dan Masyarakat.
8. Mampu bersikap profesional, etis, dan bertanggung jawab dalam menganalisis, merancang, mengimplementasikan, dan mengevaluasi solusi berbasis komputasi.
9. Mampu bekerja sama secara toleran dan kolaboratif dalam tim multidisiplin serta memiliki komitmen terhadap pembelajaran sepanjang hayat

INFORMATICS STUDY PROGRAM

Technology is currently experiencing rapid development. This rapid development can help companies become more efficient and effective in winning competition by utilizing information technology. In the industrial revolution 4.0 era, data is also growing very rapidly and is currently called big data. Big data has characteristics such as very large volume, very fast data growth, and diverse data formats. Companies can also win competition by utilizing big data.

One important profession closely related to technological and data development is informatics. In technological and data development, a combination of knowledge and skills is needed to create software that makes business processes more efficient and effective, thereby enabling competitive advantage. Additionally, a combination of knowledge and skills is needed to analyze the big data that companies possess or own. To meet these needs, informatics program graduates with strong analytical and synthesis skills are in high demand. The UKWMS Informatics Study Program was established pursuant to the Minister of Education and Culture of the Republic of Indonesia Decree No. 63/E/O/2024 dated January 10, 2024, and has been accredited in accordance with LAM INFOKOM letter No. 265/SK/LAM-INFOKOM/Ak.S/S/VI/2026.

INDEPENDENT PROFESSIONAL PROFILE

1. Able to analyze computational problems and apply computing principles and related disciplines to design, implement, and evaluate computing-based solutions appropriate to user and organizational needs.
2. Able to utilize trends and opportunities in the development of informatics or computer science technology to support innovation and organizational or community performance improvement.
3. Able to think logically, critically, and systematically in utilizing knowledge in informatics to solve real problems effectively and efficiently.
4. Able to demonstrate quality and measurable independent performance, work tolerantly and responsibly in multidisciplinary teams, and have a commitment to self-development through lifelong learning.

GRADUATE LEARNING OUTCOMES

1. Able to apply theoretical concepts of computer science including algorithms, data structures, computational mathematics, computer systems, and networks as the basis for computational problem analysis.
2. Able to apply software engineering principles, needs analysis, system design, and implementation methods and evaluation of computational solutions.
3. Able to apply current technology concepts in informatics such as data analytics, artificial intelligence, and information security to support innovation.
4. Able to think logically, critically, and systematically in identifying, formulating, and solving real computational-based problems effectively and efficiently.
5. Able to make decisions based on data and information analysis and communicate technical solutions clearly and measurably.
6. Able to analyze user and organizational needs, and design and implement computing-based solutions according to software engineering standards.
7. Able to evaluate, validate, and improve the performance of computational solutions by utilizing the latest technological developments to support organizational and community innovation.
8. Able to act professionally, ethically, and responsibly in analyzing, designing, implementing, and evaluating computing-based solutions.
9. Able to collaborate tolerantly and collaboratively in multidisciplinary teams and have a commitment to lifelong learning.

TUJUAN PENDIDIKAN

1. Menghasilkan lulusan yang tidak hanya kompeten secara teknis, tetapi juga memiliki pemahaman etika profesional, mampu beradaptasi dengan cepat terhadap perubahan teknologi, dan memberikan dampak positif dalam pengembangan dan pemanfaatan teknologi informasi di masyarakat.
2. Mengembangkan lingkungan pembelajaran dan riset yang dinamis dan kolaboratif, yang mampu mengakomodasi perkembangan terbaru dalam bidang teknologi informasi, serta menciptakan inovasi yang relevan dengan kebutuhan industri dan masyarakat dan berkelanjutan.
3. Meningkatkan kualitas kurikulum dan metode pengajaran, agar mahasiswa dapat menguasai pengetahuan dan keterampilan terkini dalam informatika dan siap untuk berkompetisi di tingkat nasional maupun internasional.
4. Menjalin kemitraan dengan industri (DUDI), pemerintah, media, masyarakat, dan akademisi lain, untuk mendukung pertukaran pengetahuan, sumber daya, dan riset bersama yang mempercepat penerapan teknologi informasi dalam memecahkan permasalahan nyata di masyarakat.
5. Mendorong implementasi nilai-nilai Pancasila dan prinsip-prinsip Katolik dalam proses pembelajaran, riset, serta kegiatan pengabdian kepada masyarakat, untuk membentuk karakter lulusan yang berintegritas dan bertanggung jawab dalam menggunakan teknologi.
6. Meningkatkan keterlibatan mahasiswa dalam proyek riset dan pengabdian masyarakat yang relevan dengan tantangan teknologi informasi, guna membekali mereka dengan pengalaman praktis dan kontribusi dan berdampak langsung kepada kesejahteraan masyarakat.

SISTEM PENDIDIKAN

Pendidikan di PS Informatika dapat ditempuh dalam waktu 8 semester (minimal 144 sks). Pemilihan topik sesuai dengan bidang minat dapat dilaksanakan pada semester ke 7 dan 8 sebanyak 15 sks dari 30 sks yang ditawarkan. Topik minat pada Program Studi Informatika adalah Data Science & Kecerdasan Buatan dan Software Engineering & Development. Mahasiswa Program Studi Informatika UKWMS memiliki fasilitas dalam menjalankan program merdeka belajar kampus merdeka (MBKM). MBKM mulai dijalankan pada semester 5 dengan memberikan kesempatan untuk menjalani program magang MBKM dan pengakuan sampai maksimal 20 SKS. Selain itu mahasiswa program magang dapat mengikuti program flagship yang diadakan oleh Kemdikristek DIKTI seperti magang dan studi independen.

KEUNGGULAN PS Informatika

1. Menghasilkan lulusan pada bidang teknologi informasi dan komunikasi yang dapat berdaya saing pada lingkungan global dengan berlandaskan nilai Katolik dan Peka, khususnya pada pengembangan perangkat Data Science.
2. Pengajar adalah akademisi yang merupakan praktisi/konsultan di bidang Informatika.
3. Bermitra dengan lembaga sertifikasi Internasional dan perusahaan (termasuk Startup) baik Nasional maupun Internasional.

PENGEMBANGAN PS Informatika

Program Studi Informatika dikembangkan ke arah pengelolaan, analisis, dan penerapan berbagai ilmu pengetahuan yang terkait dengan Big Data melalui Kecerdasan Buatan dan Data Science serta berbagai metoda/ilmu pengetahuan dan alat bantu pengembangan Perangkat Lunak (Software Development).

FASILITAS

1. Laboratorium Komputer
2. Server yang menunjang pembelajaran (antara lain Code Hosting)
3. Software berlisensi (antara lain DBeaver, Microsoft)
4. Sertifikasi Google
5. Internet & Local Area Network (LAN)
6. Wi-Fi di seluruh kampus
7. Ruang kuliah ber-AC
8. Student Lounge

EDUCATIONAL OBJECTIVES

1. **To produce graduates who are not only technically competent**, but also have an understanding of professional ethics, can adapt quickly to technological changes, and make positive impacts in the development and utilization of information technology in society.
2. **To develop a dynamic and collaborative learning and research environment**, able to accommodate the latest developments in information technology, create innovations relevant to industrial and community needs, and ensure sustainability.
3. **To improve the quality of curriculum and teaching methods**, enabling students to master up-to-date knowledge and skills in informatics and be ready to compete at national and international levels.
4. **To establish partnerships with industry, government, media, community, and other academics**, to support knowledge exchange, resources, and joint research that accelerates the application of information technology in solving real problems in society.
5. **To encourage the implementation of Pancasila values and Catholic principles** in the learning process, research, and community service activities, to shape graduates' character with integrity and responsibility in using technology.
6. **To increase student involvement in research projects and community service** relevant to information technology challenges, providing them with practical experience and contributions that directly impact community welfare.

EDUCATION SYSTEM

Education in the Informatics Study Program can be completed in 8 semesters (minimum 144 SKS). Topic selection according to specialization can be conducted in semesters 7 and 8 with 15 SKS out of 30 SKS offered. Specialization topics in the Informatics Study Program are Data Science & Artificial Intelligence and Software Engineering & Development. Informatics Study Program students have facilities to run the Independent Learning - Independent Campus (MBKM) program. MBKM starts in semester 5, providing opportunities to undertake MBKM internships that are recognized for up to 20 SKS. Additionally, internship students can participate in flagship programs organized by Kemdiktisaintek, such as internships and independent studies.

ADVANTAGES OF THE INFORMATICS STUDY PROGRAM

1. Produces graduates in the field of information and communication technology who can compete in the global environment based on Catholic values, particularly in Data Science development.
2. Lecturers are academics who are practitioners/consultants in the field of Informatics.
3. Partnerships with international certification institutions and companies (including Startups) both nationally and internationally.

INFORMATICS STUDY PROGRAM DEVELOPMENT

The Informatics Study Program is designed to support the management, analysis, and application of knowledge related to Big Data through Artificial Intelligence and Data Science, as well as methods/knowledge and software development tools (**Software Development**).

FACILITIES

1. Computer Laboratory
2. Servers supporting learning (including Code Hosting)
3. Licensed software (including DBeaver, Microsoft)
4. Google Certification
5. Internet & Local Area Network (LAN)
6. WiFi throughout the campus
7. Air-conditioned lecture rooms
8. Student Lounge

KERJA SAMA

- Taiwan Tech (d/h National Taiwan University of Science and Technology), Taiwan: Exchange Program dan Joint Degree Program.
- Beijing University of Chemical Technology: Joint Degree Program.
- Osaka Institute of Technology, Jepang: Exchange Program dan Research Internship.
- Shibaura Institute of Technology, Jepang: Exchange Program
- National Chung Cheng University, Taiwan: Exchange Program dan Research Internship.
- Zheijang University of Technology, Tiongkok: Research Internship.
- Swinburne University of Technology, Malaysia: Research Internship.
- Curtin University, Malaysia: Research Internship.
- Sophistec Global (<https://sophistec.global/>)

PROSPEK KERJA

Prospek kerja dari lulusan PS Informatika sangatlah luas baik di berbagai industri maupun organisasi antara lain:

- System Analyst/Engineer
- General Software Engineer
- Mobile Apps Developer
- Web Designer and Developer
- Database Administrator
- Network Engineer
- Cybersecurity Analyst
- Data Scientist
- Software Tester
- Penasehat dan Konsultan IT
- Software Architect
- Startup
- ML Operation
- IT mobility and Internet of Things

Email : hod-informatics@ukwms.ac.id

PARTNERSHIPS

- Taiwan Tech (formerly National Taiwan University of Science and Technology), Taiwan: Exchange Program and Joint Degree Program.
- Beijing University of Chemical Technology: Joint Degree Program.
- Osaka Institute of Technology, Japan: Exchange Program and Research Internship.
- Shibaura Institute of Technology, Japan: Exchange Program.
- National Chung Cheng University, Taiwan: Exchange Program and Research Internship.
- Zhejiang University of Technology, China: Research Internship.
- Swinburne University of Technology, Malaysia: Research Internship.
- Curtin University, Malaysia: Research Internship.
- Sophistec Global (<https://sophistec.global/>)

CAREER PROSPECTS

Career prospects for Informatics graduates are very broad in various industries and organizations including:

- System Analyst/Engineer
- General Software Engineer
- Mobile Apps Developer
- Web Designer and Developer
- Database Administrator
- Network Engineer
- Cybersecurity Analyst
- Data Scientist
- Software Tester
- IT Advisor and Consultant
- Software Architect
- Startup
- ML Operation
- IT mobility and Internet of Things

Email: hod-informatics@ukwms.ac.id

PROGRAM STUDI PROFESI INSINYUR

LATAR BELAKANG PENDIRIAN

Program Studi Profesi Insinyur Universitas Katolik Widya Mandala Surabaya (PSPI UKWMS) diselenggarakan sesuai mandat Kementerian Riset Teknologi dan Pendidikan Tinggi pada tahun 2016 untuk menjawab kebutuhan sumber daya manusia yang sesuai standar kompetensi insinyur di Indonesia. PSPI UKWMS telah terakreditasi oleh Lembaga Akreditasi Mandiri Teknik (LAMTEK) dengan predikat "UNGGUL" sesuai Surat Keputusan No. 0103/SK/LAM TEK/PI/IV/2026.

Program Profesi Insinyur UKWMS adalah kelanjutan dari pendidikan strata-1 Bidang Teknik/Terapan Bidang Teknik/Sains/Pendidikan Bidang Teknik dan memiliki kualifikasi level 7 (tujuh) dalam KKNI. Lulusan PSPI UKWMS akan mendapatkan gelar Insinyur (Ir.). Dengan memperoleh gelar Ir, maka seseorang dapat melakukan praktik keinsinyuran secara legal dan profesional.

Sesuai aturan Undang-Undang No 11 Tahun 2014 tentang Keinsinyuran bahwa seseorang yang telah menempuh Profesi Insinyur dan memiliki ijazah Profesi Insinyur maka dapat mengikuti Uji Kompetensi Insinyur Profesional yang diselenggarakan oleh Persatuan Insinyur Indonesia atau Lembaga Sertifikasi Profesi. Apabila seseorang telah lulus Uji Kompetensi Insinyur Profesional, maka akan memperoleh Sertifikat Kompetensi sebagai Insinyur Profesional sesuai dengan tingkat kompetensi dan pengalamannya yaitu Insinyur Profesional Pratama (IPP), Insinyur Profesional Madya (IPM), dan Insinyur Profesional Utama (IPU). Sertifikat kompetensi ini merupakan pengakuan kompetensi seorang insinyur untuk melakukan praktik keinsinyuran dan digunakan sebagai syarat untuk memperoleh Surat Tanda Registrasi Insinyur (STRI).

VISI MISI PROGRAM STUDI PROGRAM PROFESI INSINYUR

VISI

Program Studi Profesi Insinyur UKWMS menjadi pusat pendidikan profesi insinyur yang terbuka, diakui di tingkat nasional hingga internasional dalam bidang rekayasa keberlanjutan, untuk berkontribusi dan berdampak pada kesejahteraan masyarakat dengan dijiwai oleh nilai-nilai Pancasila dan prinsip-prinsip Katolik.

MISI

1. Melaksanakan pendidikan profesi insinyur untuk menghasilkan pelaku profesi yang andal dan berdaya saing tinggi, dengan hasil pekerjaan yang bermutu dan relevan dengan kebutuhan pembangunan berkelanjutan.
2. Melaksanakan penelitian dan pengabdian kepada masyarakat dalam bidang rekayasa berkelanjutan untuk meningkatkan kemaslahatan masyarakat.
3. Memperkuat jejaring kerja sama pentahelix untuk pengembangan Tridharma Perguruan Tinggi.

TUJUAN

1. Menghasilkan lulusan yang beretika dan profesional dalam memecahkan berbagai permasalahan/kasus keinsinyuran dengan menanamkan nilai Pancasila dan prinsip-prinsip Katolik.
2. Menghasilkan penelitian dan pengabdian pada masyarakat di bidang rekayasa berkelanjutan yang berkontribusi dan berdampak pada kemaslahatan masyarakat.
3. Memperkuat kolaborasi pentahelix yang sinergis dan berkelanjutan melalui pendidikan, penelitian dan pengabdian kepada masyarakat dalam bidang rekayasa berkelanjutan.

PROFIL LULUSAN

1. Memecahkan berbagai permasalahan/kasus dalam bidang keinsinyuran secara etis dan akuntabel dengan memperhatikan dampak kesehatan, keselamatan kerja, dan lingkungan.
2. Mengembangkan keprofesian berkelanjutan melalui karya-karya keinsinyuran yang bermutu dengan berlandaskan Pancasila, prinsip cinta kasih, dan nilai keutamaan PeKA (Peduli, Komit, dan Antusias) demi kemaslahatan masyarakat.
3. Berperan aktif dalam pembangunan nasional yang berkelanjutan dengan memajukan ilmu pengetahuan dan teknologi sesuai dengan kompetensi keinsinyuran.

PROFESSIONAL ENGINEER STUDY PROGRAM

BACKGROUND OF ESTABLISHMENT

The Professional Engineer Study Program of Widya Mandala Catholic University Surabaya (PSPI UKWMS) was established according to the mandate of the Ministry of Research, Technology, and Higher Education in 2016 to meet the need for human resources that meet Indonesian engineer competency standards. PSPI UKWMS has been accredited by the Independent Engineering Accreditation Institution (LAMTEK) with "EXCELLENT" predicate according to Decree No. 0103/SK/LAM TEK/PI/IV/2026.

The Professional Engineer Program is a continuation of undergraduate education in Engineering/Applied Engineering/Science/Engineering Education and has level 7 (seven) qualifications in KKNI. Graduates of PSPI UKWMS will receive the title of Engineer (Ir.). By obtaining the Ir title, a person can practice engineering legally and professionally.

According to Law No. 11 of 2014 concerning Engineering, a person who has completed the Professional Engineer program and holds a Professional Engineer diploma can participate in the Professional Engineer Competency Test organized by the Indonesian Engineers Association or Professional Certification Institutions. Upon passing the Professional Engineer Competency Test, one will receive a Competency Certificate as a Professional Engineer according to their competency level and experience: Pratama Professional Engineer (IPP), Madya Professional Engineer (IPM), and Utama Professional Engineer (IPU). This competency certificate recognizes an engineer's competence to practice engineering and is required to obtain the Engineer Registration Certificate (STRI).

VISION AND MISSION OF THE PROFESSIONAL ENGINEER STUDY PROGRAM

VISION

The Professional Engineer Study Program of UKWMS aspires to become an open professional engineering education center, recognized nationally and internationally in the field of sustainable engineering, contributing and impacting community welfare while being guided by Pancasila values and Catholic principles.

MISSION

1. To conduct professional engineering education that produces reliable and highly competitive professionals, with quality work relevant to sustainable development needs.
2. To conduct research and community service in the field of sustainable engineering to improve community welfare.
3. To strengthen pentahelix networking partnerships for the development of the Tri Dharma of Higher Education.

OBJECTIVES

1. To produce ethical and professional graduates capable of solving various engineering problems/cases by instilling Pancasila values and Catholic principles.
2. To produce research and community service in the field of sustainable engineering that contributes to and impacts community welfare.
3. To strengthen synergistic and sustainable pentahelix collaboration through education, research, and community service in the field of sustainable engineering.

GRADUATE PROFILES

1. Solve various engineering problems/cases ethically and accountably while considering the impact on health, occupational safety, and the environment.
2. Develop sustainable professionalism through quality engineering works based on Pancasila, the principle of love, and PeKA values (Caring, Committed, and Enthusiastic) for community welfare.
3. Actively participate in sustainable national development by advancing science and technology according to engineering competencies.

PIMPINAN PROGRAM STUDI PROFESI INSINYUR

Ketua Program Studi Program Profesi Insinyur: Dr. Ir. Ivan Gunawan, S.T., M.MT., CSCM., IPM., ASEAN Eng.

MENGAPA HARUS MEMILIH PSPI UKWMS?

- Didukung oleh dosen-dosen lulusan dari perguruan tinggi di luar dan dalam negeri yang berkualifikasi Doktor (S-3) dan Magister (S-2) serta bersertifikat minimal Insinyur Profesional Madya (IPM) serta praktisi yang ahli di bidangnya.
- Pembelajaran dilakukan secara hybrid.
- Telah terakreditasi UNGGUL oleh LAMTEK.

JALUR PROSES PEMBELAJARAN

Terdapat dua jalur proses pembelajaran, yaitu:

1. Program Reguler
2. Program Rekognisi Pembelajaran Lampau (RPL)

I. Program REGULER:

Program untuk calon peserta yang memiliki pengalaman kerja di bidang keinsinyuran kurang dari 2 tahun. Pembelajaran dilaksanakan selama dua semester mencakup pertemuan secara daring, FGD, pelaksanaan tugas mandiri atau kelompok.

Syarat khusus pendaftaran:

- Sarjana Bidang Teknik/Terapan Bidang Teknik/Sains/Pendidikan Bidang Teknik.
- Untuk Sarjana Sains/Pendidikan Bidang Teknik dengan pengalaman kerja di bidang keinsinyuran minimal 3 tahun.

Lama dan Beban Studi

- Lama studi 2 semester
- Beban studi 36 sks

II. Program REKOGNISI PEMBELAJARAN LAMPAU (RPL):

Program RPL adalah pengakuan terhadap Capaian Pembelajaran yang diperoleh dari pengalaman kerja pada bidang keinsinyuran. Pengakuan pengalaman tersebut melalui dokumen portofolio dan wawancara (tanpa perkuliahan).

Syarat khusus pendaftaran:

- Sarjana Bidang Teknik/Terapan Bidang Teknik
- Telah memiliki pengalaman kerja pengalaman kerja pada bidang keinsinyuran lebih dari 2 (dua) tahun.

Lama dan Beban Studi

- Lama studi 1-2 semester
- 36 sks

KURIKULUM

1. Kurikulum PSPI dituangkan dalam mata kuliah sebagai berikut:
2. Kode Etik dan Etika Profesi Insinyur (4 sks)
3. Profesionalisme (4 sks)
4. Keselamatan, Kesehatan dan Keamanan Kerja dan Lingkungan (4 sks)
5. Studi Kasus (6 sks)
6. Materi pada Seminar, Workshop, dan Diskusi (4 sks)

Praktik Kerja Keinsinyuran (14 sks) meliputi:

- Filosofi Keinsinyuran di Industri
- Arah Perkembangan Industri dan Status
- Sistem Industri (Engineering)

HEAD OF THE PROFESSIONAL ENGINEER STUDY PROGRAM

Head of the Professional Engineer Study Program: Dr. Ir. Ivan Gunawan, S.T., M.MT., CSCM., IPM., ASEAN Eng.

WHY CHOOSE PSPI UKWMS?

- Supported by lecturers graduated from domestic and foreign universities with Doctoral (S-3) and Master's (S-2) qualifications, certified at minimum Madya Professional Engineer (IPM), and practitioners expert in their fields.
- Learning is conducted hybrid.
- Accredited EXCELLENT by LAMTEK.

LEARNING PATHWAYS

There are two learning pathways:

1. Regular Program
2. Recognition of Prior Learning (RPL) Program

I. REGULAR PROGRAM:

Program for prospective participants with less than 2 years of engineering work experience. Learning is conducted over two semesters, including online meetings, FGDs, and individual or group assignments.

Special registration requirements:

- Bachelor's degree in Engineering/Applied Engineering/Science/Engineering Education.
- For Bachelor of Science/Engineering Education with minimum 3 years of engineering work experience.

Study Duration and Load:

- Study duration: 2 semesters
- Study load: 36 SKS

II. RECOGNITION OF PRIOR LEARNING (RPL) PROGRAM:

The RPL program recognizes Learning Outcomes obtained from work experience in the engineering field. Recognition is based on portfolio documents and interviews (without lectures).

Special registration requirements:

- Bachelor's degree in Engineering/Applied Engineering
- Has more than 2 years of engineering work experience.

Study Duration and Load:

- Study duration: 1-2 semesters
- 36 SKS

CURRICULUM

1. The PSPI curriculum includes the following courses:
2. Code of Ethics and Engineering Professional Ethics (4 SKS)
3. Professionalism (4 SKS)
4. Occupational Safety, Health, Security, and Environment (4 SKS)
5. Case Studies (6 SKS)
6. Speaker at Seminars, Workshops, and Discussions (4 SKS)

Engineering Work Practice (14 SKS) including:

- Engineering Philosophy in Industry
- Direction of Industrial Development and Status
- Industrial Systems (Engineering)

- Permasalahan Keinsinyuran
- tugas Mengatasi Masalah
- Penulisan Praktik Keinsinyuran

SYARAT PENDAFTARAN

1. Salinan Ijazah S1
2. Salinan Transkrip S1
3. Surat Keterangan Pengalaman Kerja dari Instansi/Perusahaan
4. Salinan Kartu Tanda Penduduk (KTP)
5. Surat Keterangan Sehat
6. Pas Foto terbaru ukuran 4x6 (Berwarna dengan Latar Belakang Putih)

Bukti pembayaran biaya pendaftaran/seleksi sebesar Rp 500.000 ke Bank Mandiri, No Rekening: 1410021432471 a.n. Yayasan Wdya Mandala Surabaya Fakultas Teknik.

JADWAL PENDAFTARAN dan SELEKSI

Periode Semester Gasal Pendaftaran dan Seleksi : Januari- Agustus Perkuliahan : Agustus*

Periode Semester Genap Pendaftaran dan Seleksi : September-Januari Perkuliahan : Januari*

*) Menyesuaikan dengan kalender akademik

INFO LEBIH LANJUT

Program Studi Profesi Insinyur Jalan Kalijudan 37 Surabaya HP/WA: 0818507566

Email: pspipi@ukwms.ac.id

Admission Center UKWMS Jalan Dinoyo 42-44 Surabaya Telp: (031) 5678478 ext 141

WA: 08785228488;

Email: wmpmb@ukwms.ac.id

- Engineering Problems
- Problem-solving Assignments
- Engineering Practice Writing

REGISTRATION REQUIREMENTS

1. Copy of Bachelor's Diploma
2. Copy of Bachelor's Transcript
3. Work Experience Certificate from Institution/Company
4. Copy of Identity Card (KTP)
5. Health Certificate
6. Recent 4x6 photo (Color with White Background) Proof of registration/selection fee payment of Rp 500,000 to Bank Mandiri, Account Number: 1410021432471 a.n. Yayasan Widya Mandala Surabaya Fakultas Teknik.

REGISTRATION AND SELECTION SCHEDULE

Odd Semester Period Registration and Selection: January – August

Lectures: August* Even Semester Period Registration and Selection: September – January Lectures: January*

*) Adjusted to the academic calendar

FURTHER INFORMATION

Professional Engineer Study Program

Jl. Kalijudan 37 Surabaya

HP/WA: 0818507566

Email: pspipi@ukwms.ac.id

Admission Center UKWMS

Jl. Dinoyo 42-44 Surabaya

Telp: (031) 5678478 ext 141

WA: 08785228488

Email: wmpmb@ukwms.ac.id

PROGRAM STUDI MAGISTER TEKNIK KIMIA

Kebutuhan tenaga kerja profesional yang berkualitas mulai menjadi kebutuhan berbagai perusahaan nasional maupun internasional. Salah satu indikator berkualitas tidaknya sumber daya manusia (SDM) adalah berdasarkan strata pendidikan. Kemampuan profesional yang lebih komprehensif dan mendalam terutama pada bidang keilmuan Teknik Kimia berdasarkan pengembangan teknologi terkini dan perancangan teknologi masa depan dapat dipelajari di Program Studi Magister Teknik Kimia (PS-MTK) ini. Keunggulan dari PS-MTK ini adalah adanya 4 bidang minat yaitu: (1) Teknologi Hijau Berkelanjutan, (2) Bioteknologi dan Rekayasa Hayati, (3) Teknologi Material dan Proses, dan (4) Inovasi dan Manajemen Pabrik yang saat ini juga menjadi fokus dari pemerintah dalam rangka Revolusi Industri 4.0 dan Society 5.0 yang berbasis perkembangan sains dan teknologi. PS-MTK UKWMS ini berstatus terakreditasi dengan Peringkat Baik Sekali berdasarkan SK LAM TEKNIK Nomor 0252/SK/LAM Teknik/AM/VIII/2023.

CAPAIAN PEMBELAJARAN LULUSAN DARI PS-MTK

1. Mampu memahami dan menerapkan prinsip etika profesional serta tanggung jawab sosial dalam praktik teknik kimia, termasuk dampak lingkungan dan keberlanjutan.
2. Mampu menguasai pengetahuan lanjut di bidang teknik kimia, termasuk prinsip-prinsip dasar, teori, dan aplikasi terkini dalam proses kimia, termodinamika, kinetika reaksi, dan teknologi pemrosesan bahan.
3. Mampu merancang, melaksanakan, dan mengelola penelitian independen untuk menghasilkan inovasi di bidang teknik kimia, termasuk pengembangan produk, proses, atau teknologi baru.
4. Mampu menganalisis dan memecahkan masalah kompleks di bidang teknik kimia melalui pendekatan multidisiplin, termasuk penggunaan metode numerik, simulasi, dan eksperimen.

TUJUAN PENDIDIKAN PS-MTK

1. Menghasilkan Magister yang memiliki kemampuan penelitian mumpuni, profesional dan berjiwa entrepreneur sebagai research ataupun process engineer untuk dapat bersaing secara global.
2. Menerapkan sistem pembelajaran yang life-long learning dengan menerapkan Tridarma Perguruan Tinggi.
3. Mampu melaksanakan penelitian dan mempublikasikannya pada skala nasional dan internasional untuk pengembangan ilmu pengetahuan dan pengabdian kepada masyarakat.
4. Membangun network dengan pemangku kepentingan dan lembaga-lembaga lainnya dalam peningkatan relevansi pembelajaran dengan pemenuhan kebutuhan industri profesional dan teknologi.

SISTEM PENDIDIKAN

Pendidikan di PS-MTK ini, secara umum dapat diselesaikan dalam waktu 4 semester (54 sks) dimana perkuliahan akan dilaksanakan selama 2 semester sedangkan penelitian dengan topik sesuai dengan bidang minat dapat dilaksanakan pada semester 3, dan sidang tesis di semester 4. Selain itu, dalam peningkatan wawasan keilmuan, pengalaman lapangan dan keterampilan, mahasiswa PS-MTK diwajibkan untuk menghadiri seminar grup maupun seminar nasional, dan publikasi ilmiah. Selain jalur regular, terdapat dua jalur lain, yaitu melalui program bundling dan program percepatan. Kedua program ini adalah untuk mahasiswa S2 Program Studi Teknik Kimia UKWMS yang berencana lulus dari S1 dalam 7 semester (program bundling) dan 8 semester (program percepatan) dan langsung melanjutkan studi S2 di PS-MTK UKWMS.

Syarat khusus pendaftaran, beban studi dan lama studi untuk program:

1. Reguler

- a. Sarjana bidang Teknik Kimia atau sejenis
- b. Memiliki IPK $\geq 2,50$
- c. Lama studi 4 semester dengan beban studi sebesar 54 SKS

2. Bundling dan percepatan

- a. Mahasiswa S1 Program Studi Teknik Kimia UKWMS
- b. Memiliki IPK pada semester 6 $\geq 3,50$ (program bundling) dan IPK pada semester 7 $\geq 3,50$ (program percepatan)
- c. Lama studi 3 semester dengan beban studi sebesar 54 SKS

MASTER'S PROGRAM IN CHEMICAL ENGINEERING

The need for quality professional workers has become a requirement for various national and international companies. One indicator of the quality of human resources is based on their education level. Comprehensive and in-depth professional abilities, especially in the field of Chemical Engineering based on current technology development and future technology design, can be studied in this Master's Program in Chemical Engineering (PS-MTK). The advantages of this program include 4 fields of specialization: (1) Sustainable Green Technology, (2) Biotechnology and Biological Engineering, (3) Material and Process Technology, and (4) Innovation and Factory Management, which are also currently the focus of the government in the context of Industrial Revolution 4.0 and Society 5.0 based on science and technology development. PS-MTK UKWMS is accredited with "Very Good" status based on LAM TEKNIK Decree Number 0252/SK/LAM Teknik/AM/VIII/2023.

GRADUATE LEARNING OUTCOMES OF PS-MTK

1. Able to understand and apply professional ethical principles and social responsibility in chemical engineering practice, including environmental impact and sustainability.
2. Able to master advanced knowledge in chemical engineering, including basic principles, theories, and current applications in chemical processes, thermodynamics, reaction kinetics, and materials processing technology.
3. Able to design, conduct, and manage independent research to produce innovations in chemical engineering, including product development, processes, or new technologies.
4. Able to analyze and solve complex problems in chemical engineering through multidisciplinary approaches, including the use of numerical methods, simulation, and experimentation.

EDUCATIONAL OBJECTIVES OF PS-MTK

1. To produce Master's graduates with excellent research capabilities, a professional and entrepreneurial spirit as research or process engineers able to compete globally.
2. To implement lifelong learning systems by applying the Tri Dharma of Higher Education.
3. To conduct research and publish it nationally and internationally for the development of science and community service.
4. To build networks with stakeholders and other institutions in improving the relevance of learning to meet professional and technological industry needs.

EDUCATION SYSTEM

Education in PS-MTK can generally be completed in 4 semesters (54 SKS), with lectures conducted over 2 semesters, research aligned with the specialization in semester 3, and thesis defense in semester 4. Additionally, to enhance scientific insight, field experience, and skills, PS-MTK students are required to attend group and national seminars and to publish scientific work. Besides the regular pathway, there are two other pathways: through the bundling program and the acceleration program. Both programs are for Master's students in Chemical Engineering who plan to complete undergraduate studies in 7 semesters (bundling program) or 8 semesters (acceleration program) and immediately continue to Master's studies at PS-MTK UKWMS.

Special registration requirements, study load, and study duration:

1. Regular Program

- a. Bachelor's degree in Chemical Engineering or related field
- b. Minimum GPA ≥ 2.50
- c. Study duration: 4 semesters with a study load of 54 SKS

2. Bundling and Acceleration Program

- a. Undergraduate student in Chemical Engineering UKWMS
- b. GPA in semester 6 ≥ 3.50 (bundling program) and GPA in semester 7 ≥ 3.50 (acceleration program)
- c. Study duration: 3 semesters with a study load of 54 SKS

3. Riset

- a. Sarjana bidang Teknik Kimia atau sejenis
- b. Memiliki IPK $\geq 2,75$
- c. Lama studi 4 semester dengan beban studi sebesar 54 SKS
- d. Surat rekomendasi dari dosen prodi

KEUNGGULAN PS-MTK

1. Adanya program bundling dan program percepatan bagi mahasiswa S1 Program Studi Teknik Kimia UKMWS yang memungkinkan program magister ditempuh hanya dalam 3 semester (lulus 1,5 tahun)*
2. Bekerjasama dengan Surya Edukasi Bangsa Foundation memberikan beasiswa S2 melalui program Online Scholarship Competition (OSC)
3. Penerapan proses Life Long Learning
4. Bidang minat sesuai dengan kebutuhan industri dan perkembangan IPTEK
5. Seluruh dosen bergelar S-3 dari dalam dan luar negeri
6. Dapat melakukan kolaborasi riset/magang dengan universitas partner di luar negeri*

* Sesuai dengan ketentuan Program Studi

STRATEGI PENGEMBANGAN PS-MTK

1. Pengembangan kurikulum melalui kegiatan evaluasi, tracer study alumni dan pengguna lulusan.
2. Peningkatan akses pendidikan yang berbasis pada kualitas calon mahasiswa melalui kegiatan promosi.
3. Peningkatan jumlah kerjasama dan jejaring kerjasama di bidang pendidikan dan riset dengan institusi di dalam dan luar negeri.
4. Peningkatan kualitas penelitian dosen dengan melibatkan mahasiswa melalui pembelajaran berbasis proyek dengan implementasi student centered learning (SCL).
5. Peningkatan dampak sosial yang baik melalui kegiatan pengabdian kepada masyarakat dalam negeri.

FASILITAS

1. Laboratorium :
 - Analisa Instrumen
 - Kimia Organik dan Kimia Fisika
 - Kimia Analisa
 - Operasi Teknik Kimia
 - Teknologi Bioproses
 - Teknologi Proses
 - Komputer Terpadu
2. Internet & Local Area Network (LAN)
3. Ruang kuliah ber-AC
4. Ruang baca jurusan
5. Wi-fi

KERJA SAMA

1. National Taiwan University of Science and Technology, Taiwan (Penelitian)
2. Osaka Institute Technology (OIT), Jepang (Magang, Penelitian)
3. Shibaura Institute of Technology (SIT), Jepang (International Joint Activities)
4. Perbanusa DPD I Jatim (Pengabdian kepada masyarakat)
5. LSM, Desa Morang, Kecamatan Kare, Kabupaten Madiun (Pengabdian kepada masyarakat)

3. Research Program

- a. Bachelor's degree in Chemical Engineering or related field
- b. Minimum GPA ≥ 2.75
- c. Study duration: 4 semesters with a study load of 54 SKS
- d. Recommendation letter from study program lecturer

ADVANTAGES OF PS-MTK

1. Bundling program and acceleration program for S1 Chemical Engineering UKWMS students, allowing the master's program to be completed in only 3 semesters (graduate in 1.5 years)*
2. Collaboration with Surya Edukasi Bangsa Foundation providing S2 scholarships through the Online Scholarship Competition (OSC) program
3. Implementation of lifelong learning processes
4. Specialization fields aligned with industrial needs and science and technology development
5. All lecturers hold S-3 degrees from domestic and foreign universities
6. Opportunities for research collaboration/internships with partner universities abroad*

*) Subject to Study Program provisions

PS-MTK DEVELOPMENT STRATEGY

1. Curriculum development through evaluation activities, alumni tracer studies, and graduate user feedback.
2. Improving access to education based on the quality of prospective students through promotional activities.
3. Increasing the number of partnerships and networking in education and research with domestic and foreign institutions.
4. Improving the quality of lecturer research involving students through project-based learning with student-centered learning (SCL) implementation.
5. Improving positive social impact through community service activities.

FACILITIES

1. Laboratories:
 - Instrument Analysis
 - Organic Chemistry and Physical Chemistry
 - Analytical Chemistry
 - Chemical Engineering Operations
 - Bioprocess Technology
 - Process Technology
 - Integrated Computer
2. Internet & Local Area Network (LAN)
3. Air-conditioned lecture rooms
4. Department reading room
5. Wi-fi

PARTNERSHIPS

1. National Taiwan University of Science and Technology, Taiwan (Research)
2. Osaka Institute Technology (OIT), Japan (Internship, Research)
3. Shibaura Institute of Technology (SIT), Japan (International Joint Activities)
4. Perbanusa DPD I Jatim (Community Service)
5. NGOs, Morang Village, Kare District, Madiun Regency (Community Service)

PROSPEK KERJA

Prospek kerja dari lulusan PS-MTK sangatlah luas baik di industri kimia, minyak/gas, pangan, kosmetik, maupun pendidikan sebagai Environmental Risk Assessor; Environmental, Safety & Health Engineer; peneliti; dosen; Staff Research and Development; Project Engineer; Process Engineer; Management Consultant; Engineering Consultant; Process Safety Engineer; Liaison Engineer; Analyst; Quality Control/Assurance Engineer; dll.

Email : hod-chemengmaster@ukwms.ac.id

CAREER PROSPECTS

Career prospects for PS-MTK graduates are very broad in chemical industries, oil/gas, food, cosmetics, and education as Environmental Risk Assessor, Environmental, Safety & Health Engineer, Researcher, Lecturer, Staff Research and Development, Project Engineer, Process Engineer, Management Consultant, Engineering Consultant, Process Safety Engineer, Liaison Engineer, Analyst, Quality Control/Assurance Engineer.

Email: hod-chemengmaster@ukwms.ac.id

